

Orthogonal Dimensionality Reduction for Feature Descriptors: A Denoising Perspective on Compact Representations

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Abstract: High-dimensional feature descriptors impose substantial computational and memory burdens that limit their deployment in resource-constrained computer vision applications. While dimensionality reduction techniques such as Principal Component Analysis (PCA) are widely employed, they optimize global variance criteria that can suppress discriminative structure essential for vision tasks. We introduce Gram-Schmidt Feature Reduction (GFR), a sequential orthogonalization method that iteratively identifies principal directions in residual variance while enforcing orthogonality in the sample-score space. By systematically decomposing the descriptor space and preserving complementary discriminative components, GFR yields compact representations that maintain or enhance task performance compared to full-dimensional descriptors. We evaluate GFR across visual localization benchmarks (Aachen Day-Night, InLoc) and patch-level tasks (HPatches) using both hand-crafted (SIFT) and learned (HardNet) descriptors. Results demonstrate that GFR-compressed descriptors at 16–64 dimensions consistently match or exceed 128-dimensional baselines while achieving 2–4× matching speedup and 50–75% storage reduction. Notably, GFR-64 attains 88.2% localization accuracy on Aachen Day-Night compared to 87.2% for SIFT-128, suggesting an effective denoising mechanism that filters redundant information while retaining discriminative power. Our method requires no descriptor-specific training and generalizes across descriptor paradigms, enabling practical deployment in memory-constrained robotics and real-time vision systems. These findings establish iterative orthogonalization as a principled alternative to variance-maximization approaches for descriptor compression.

1 INTRODUCTION

The exponential growth of visual data demands efficient storage, transmission, and processing of image features. High-dimensional local feature descriptors, while fundamental to computer vision tasks, impose significant computational burdens that limit deployment in resource-constrained environments and real-time applications. High-dimensional feature spaces incur substantial costs in both computation and memory. For example, matching n_q query descriptors against n_d database descriptors of dimension d requires $O(n_q \times n_d \times d)$ time and $O(N \times d)$ storage for N descriptors.

Contemporary feature descriptors span two major paradigms: hand-crafted methods like SIFT (Lowe,

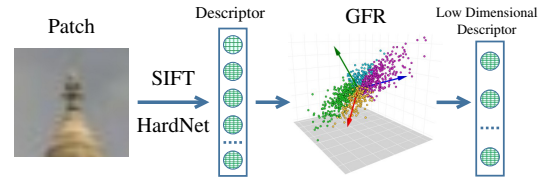


Figure 1: Overview of our approach. First, descriptors are computed from image patches. Then, Gram-Schmidt feature reduction is applied for dimensionality reduction. Finally, the reduced descriptors are used in various vision tasks.

2004), SURF (Bay et al., 2006), BRIEF (Calonder et al., 2012), and ORB (Rublee et al., 2011), and learning-based techniques such as TFeat (Balntas et al., 2016), HardNet (Mishchuk et al., 2017), L2-Net (Tian et al., 2017a), and SOSNet (Tian et al., 2019). Classical methods employ domain-specific transformations optimized for robustness to geometric and photometric variations, while learning-based approaches leverage large datasets (Brown et al., 2011) and specialized loss

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functions (Luo et al., 2018; Ebel et al., 2019; Han et al., 2015; Simo-Serra et al., 2015; Tian et al., 2017b) to achieve superior discriminative performance. Despite their robustness, their typical 128-512 dimensional representations create bottlenecks in feature matching (quadratic complexity), database storage (linear growth with collection size), and memory-constrained mobile robotics applications (Fergus et al., 2003; Mikolajczyk and Schmid, 2001). Thus, effective dimensionality reduction techniques are essential to alleviate these computational and storage bottlenecks while maintaining descriptor discriminativeness in diverse vision applications.

Prior descriptor compression efforts span feature selection strategies (Mao, 2004; Breiman, 2001) and linear projection techniques dominated by PCA (Jolliffe, 2002a; Jolliffe, 2002b), factor analysis (Watkins, 2018), and LDA (Balakrishnan and Ganapathiraju, 1998). Applied to local features, these methods (e.g., PCA for SIFT and SURF (Valenzuela et al., 2014; Gil et al., 2006) or PCA-SIFT on raw patches (Ke and Sukthankar, 2004a; Ke and Sukthankar, 2004b)) primarily optimize global variance and often suppress discriminative structure that underpins the robustness highlighted above. Non-linear alternatives such as Kernel PCA (Schölkopf et al., 1997), t-SNE (van der Maaten and Hinton, 2008), and isometric mapping (Rassias, 1999) capture richer relationships but incur prohibitive computational costs for real-time deployment. Recently, Dong *et al.* (Dong et al., 2022) propose an MLP auto-encoder trained with reconstruction and distance-preservation losses to learn nonlinear compact projections that outperform PCA across descriptors. However, Dong et al.’s MLP auto-encoder requires descriptor-specific training and expensive retraining when deployment conditions change, which limits its practicality.

To address these limitations, we introduce Gram-Schmidt Feature Reduction (GFR), a GS-sequential linear method that iteratively fits principal directions on residuals and orthogonalizes the resulting score vectors. By enforcing orthogonality in the sample-score space and selecting directions from residual variance, GFR preserves complementary discriminative components that PCA’s global criterion can suppress, yielding compact, denoised descriptors for both hand-crafted (SIFT) and learned (HardNet) representations.

To validate this approach, we conducted comprehensive evaluation across visual localization (Aachen Day-Night, InLoc), patch-level tasks (HPatches), and two descriptor types (SIFT, HardNet), demonstrating that GFR-compressed descriptors often match or slightly exceed their full-dimensional counterparts’

performance. This result suggests that dimensionality reduction acts as an effective denoising mechanism, filtering redundant information while preserving essential discriminative components. Our method reduces descriptor dimensionality from 128 to 16, 24, 32, or 64 dimensions while maintaining or improving task performance, enabling deployment in resource-constrained scenarios.

Our key contributions are: (i) a Gram-Schmidt feature reduction procedure that iteratively fits residual principal directions and orthogonalizes descriptor scores to preserve complementary discriminative information; (ii) an analysis of PCA’s limitations together with ablation and complexity studies that clarify when GS-sequential projections yield denoising benefits; (iii) a comprehensive evaluation on SIFT and HardNet across Aachen Day-Night, InLoc, and HPatches tasks showing that 16–64D GFR descriptors match or surpass 128D baselines; and (iv) evidence of 2–8× memory and 2–4× matching speed savings that enable deployment in resource-constrained visual localization pipelines.

2 Methodology

Our methodology formalizes Gram-Schmidt Feature Reduction (GFR) as a sequential augmentation of PCA that retains discriminative structure while improving efficiency. We first summarize the end-to-end pipeline (Figure 2), then detail the orthogonalization mechanics and iterative reduction procedure that yield compact descriptors.

2.1 GFR Framework Overview

The GFR workflow for descriptor dimensionality reduction proceeds as follows (Figure 2):

1. **Descriptor Extraction:** Extract local descriptors $\{d_i\}_{i=1}^N$ from image patches using SIFT or HardNet, where each descriptor $d_i \in \mathbb{R}^{128}$.
2. **Second-Order Gram-Schmidt Expansion:** Form first-order PCA scores and their second-order interaction z_{12} , orthogonalizing each against previously accepted components to build the augmented score basis.
3. **Projection and Normalization:** Project descriptors onto each candidate direction and ℓ_2 -normalize the resulting scores to obtain orthonormal components.
4. **Residual Update:** Subtract the contribution of accepted components from the descriptor matrix

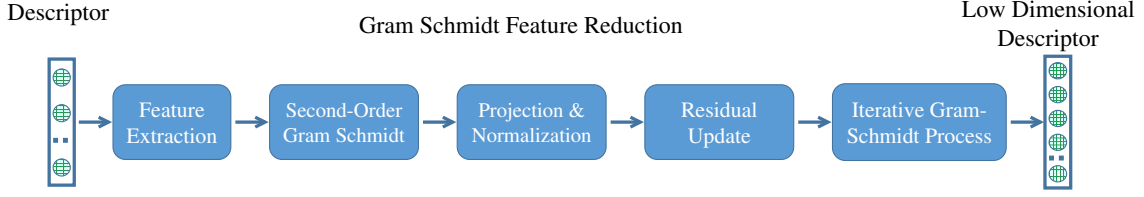


Figure 2: Overview of the Gram-Schmidt feature reduction pipeline: descriptors are extracted, expanded with second-order Gram-Schmidt interactions, projected and ℓ_2 -normalized, used to update the residual, and iteratively refined until convergence.

so that subsequent PCA operates on unexplained variance.

5. **Iterative GS Refinement:** Repeat projection, orthogonalization, higher-order augmentation, and residual updates while eigenvalues exceed ϵ , yielding the final projection matrix V .

In what follows, we first describe the foundational orthogonalization algorithm before detailing how it is employed within the GS-sequential procedure to generate compact, orthogonal descriptor representations.

2.2 Gram-Schmidt Orthogonalization: Core Mechanism

The foundation of our method is the Gram-Schmidt process, which we use to systematically decompose the high-dimensional descriptor space. This process is formalized in Algorithm 1. At each iteration j , the algorithm takes a new projected component z_j and makes it orthogonal to all previously computed components $\{\hat{z}_i\}_{i=1}^{j-1}$. This is achieved by subtracting its projection onto the already-orthogonalized components (Line 1) and then normalizing the result (Line 1).

This process can be concisely represented by the classical Gram-Schmidt projection formula. For a new component z_j and a set of existing orthonormal components $\{\hat{z}_i\}_{i=1}^{j-1}$, the orthogonalization is:

$$\tilde{z}_j = z_j - \sum_{i=1}^{j-1} \langle z_j, \hat{z}_i \rangle \hat{z}_i, \quad \text{and} \quad \hat{z}_j = \frac{\tilde{z}_j}{\|\tilde{z}_j\|} \quad (1)$$

where $\tilde{z}_j$ is the orthogonalized component and $\hat{z}_j$ is its normalized version; $\langle \cdot, \cdot \rangle$ denotes the standard inner product over sample scores in $\mathbb{R}^N$. A critical step in Algorithm 1 is updating the residual data matrix d_j by removing the contributions of all orthogonalized components from the original data. This residual matrix captures the variance structure *after* the information from the first j components has been removed, which is essential for finding the next most important feature direction through PCA.

Input:

- z_j : new projected component from eigenvector v_j
- $\{\hat{z}_i\}_{i=1}^{j-1}$: previously orthogonalized score components
- $d_{j-1} \in \mathbb{R}^{d \times N}$: residual data matrix at step $j-1$
- $v_j \in \mathbb{R}^d$: current projection direction

Output:

- $\hat{z}_j$: orthogonalized and normalized component
- d_j : updated residual data matrix

Initialize: $\tilde{z}_j \leftarrow z_j$.

for $i \leftarrow 1$ **to** $j-1$ **do**

$\tilde{z}_j \leftarrow \tilde{z}_j - \langle \tilde{z}_j, \hat{z}_i \rangle \hat{z}_i$;

end

$\hat{z}_j \leftarrow \frac{\tilde{z}_j}{\|\tilde{z}_j\|}$.

Update residual data:

$d_j \leftarrow d_{j-1} - v_j(v_j^T d_{j-1})$.

Return $\hat{z}_j, d_j$.

Algorithm 1: Orthogonalize($z_j, \{\hat{z}_i\}_{i=1}^{j-1}, d_{j-1}, v_j$)

2.3 Descriptor Dimensionality Reduction with GFR

We operationalize the orthogonalization process using Gram-Schmidt Feature Reduction (GFR), as detailed in Algorithm 2, to produce compact descriptor representations.

The GFR algorithm iteratively identifies orthogonal components that maximize variance in the descriptor space. At each step j , it applies PCA to the residual data matrix d_{j-1} to find the principal eigenvector v_j and eigenvalue λ_j (Line 2). This eigenvector represents the direction of maximum variance in the descriptor information that has not yet been captured by previous components.

Primary Component (v_1): The first component is generated by applying PCA to the centered data d_0 to find the principal eigenvector v_1 . This vector captures the direction of maximum variance in the original 128-dimensional descriptor space:

$$z_1 = v_1^T d_0 \quad (2)$$

Input: Data matrix $X \in \mathbb{R}^{128 \times N}$ (columns are descriptors), target dimension $m \in \{16, 24, 32, 64\}$, threshold $\epsilon > 0$.
Output: Projection matrix $V \in \mathbb{R}^{128 \times m}$ and reduced descriptors $\tilde{X} \in \mathbb{R}^{m \times N}$.
Initialize: mean-center X to obtain d_0 , set $V \leftarrow \emptyset, \hat{z} \leftarrow \emptyset, j \leftarrow 1$.
while $j \leq m$ **do**
 Apply PCA to d_{j-1}^T to get top eigenvector v_j and eigenvalue λ_j .
 if $\lambda_j \leq \epsilon$ **then**
 break.
 end
 $z_j \leftarrow v_j^T d_{j-1}$.
 $\hat{z}_j, d_j \leftarrow$
 Orthogonalize($z_j, \{\hat{z}_i\}_{i=1}^{j-1}, d_{j-1}, v_j$).
 $V \leftarrow [V, v_j]$.
 $\hat{z} \leftarrow [\hat{z}, \hat{z}_j]$.
 $j \leftarrow j + 1$.
end
Generate reduced descriptors: $\tilde{X} \leftarrow V^T X$.
Return $V, \tilde{X}$

Algorithm 2: Gram–Schmidt (GS) Sequential Reduction for Descriptor Dimensionality

Subsequent Components (v_j): Beyond the primary component, subsequent directions are generated by iteratively applying PCA to the residual matrix d_{j-1} after removing the contributions of previous components. For each new component v_j , we project from the current residual and then orthogonalize the scores:

$$z_j = v_j^T d_{j-1}, \quad \text{followed by} \quad \hat{z}_j = \text{Orthogonalize}(z_j, \{\hat{z}_i\}_{i=1}^{j-1}, d_{j-1}, v_j) \quad (3)$$

The final reduced descriptor matrix is obtained by projecting the original data matrix onto the selected eigenvectors:

$$\tilde{X} = V^T X \quad (4)$$

where $V = [v_1, v_2, \dots, v_m] \in \mathbb{R}^{128 \times m}$ is the projection matrix containing the $m \in \{16, 24, 32, 64\}$ most significant eigenvectors.

This iterative process continues until the desired number of components m is reached or the eigenvalue falls below the threshold ϵ , providing a compact representation that preserves the most discriminative information from the original high-dimensional descriptors while ensuring orthogonality between components. When Step 2 selects the top-variance direction on the residual, this GS-sequential procedure is equivalent to classical PCA with deflation; empirically observed gains stem from centering/normalization choices and the denoising effect of reduced dimensionality.

3 Experimental Validation

This section systematically evaluates Gram–Schmidt feature reduction (GFR) across visual localization, matching, retrieval, and verification benchmarks while contrasting its accuracy, runtime, and memory usage against PCA, KPCA, and deep unsupervised baselines.

3.1 Experimental Design

Descriptor Selection. We evaluate two representative descriptor types: SIFT (Lowe, 2004), a hand-crafted, gradient-based descriptor, and HardNet (Mishchuk et al., 2017), a learned CNN-based descriptor.

Dimensionality Configurations: We compress 128-D descriptors to 64, 32, 24, and 16 dimensions (2:1–8:1) to study performance vs. efficiency trade-offs.

Evaluation Framework: Tasks include Visual Localization (Aachen Day–Night v1.1 (Toft et al., 2022), InLoc (Taira et al., 2018)) and patch-level tasks on HPatches (Balntas et al., 2017) (matching, retrieval, verification).

Baseline Comparisons: We compare GFR to PCA, Kernel PCA (KPCA), and unsupervised deep learning (DL-US) to cover linear, non-linear, and deep-unsupervised reduction strategies.

Implementation Details: Localization uses a hierarchical pipeline (Sarlin et al., 2019); success is measured at 0.25m/2°, 0.5m/5°, and 5m/10°. HPatches follows standard matching/retrieval/verification protocols (mAP). Matching uses L2 distance, Lowe ratio with mutual check, and PnP+RANSAC for pose estimation.

Training Protocol and Data Splits: Projection matrices and baselines are trained on dedicated training data; evaluation queries from Aachen, InLoc, and HPatches are excluded. Hyperparameters are selected on validation splits.

Runtime and Memory Profiling: We report per-query matching time, index build time, and memory footprint for 128/64/32/24/16-D descriptors using an ANN backend (e.g., FAISS/HNSW), presenting results as functions of descriptor dimension d .

3.2 Visual Localization: Validating Discriminative Preservation

Visual localization provides the most stringent test of our theoretical claims, as successful camera pose estimation requires precise preservation of discriminative spatial relationships. We evaluate both outdoor (Aachen) and indoor (InLoc) scenarios to assess robustness across environmental conditions.

Dim	Method	InLoc DUC2	Aachen Day	Aachen Night
128	SIFT	26.1/39.4/49.3	87.2/93.7/97.4	64.2/76.7/91.2
16	PCA	15.4/26.2/31.3	83.7/89.8/94.9	37.8/46.6/58.0
	DL-US	17.4/26.9/31.6	83.7/ 91.5 /95.0	35.2/44.6/ 60.1
	K-PCA	19.7 /28.5/33.3	84.2/90.0/94.6	33.7/41.5/57.0
	GFR	19.6/ 31.5/36.8	85.0 /90.6/ 95.2	34.2/45.6/58.0
24	PCA	19.6/30.1/37.8	85.3/92.3/ 96.5	49.8/60.1/76.7
	DL-US	22.6/34.5/41.8	85.9/92.4/96.2	49.8/ 63.8/77.7
	K-PCA	21.5/34.2/40.3	86.8/92.5/96.3	48.2/59.1/74.2
	GFR	25.8/38.5/46.1	87.0/92.9 /96.3	50.8 /62.2/ 77.7
32	PCA	24.6/34.7/44.3	86.2/92.1/96.6	55.9/68.4/79.8
	DL-US	22.6/36.1/44.6	86.1/93.2/96.8	57.5 /70.5/ 82.4
	K-PCA	27.3/38.0/46.4	87.0/92.9/97.1	56.5/65.3/81.9
	GFR	28.4/40.3/48.9	87.8/93.5/97.3	57.5 /71.0/81.9
64	PCA	24.1/36.0/46.3	86.2/93.4/97.2	59.6/73.6/87.6
	DL-US	28.7/40.3/50.6	86.7/ 94.3 /97.5	65.3/78.8 /89.7
	K-PCA	27.0/41.2/49.7	87.0/93.9/97.2	59.6/75.1/88.1
	GFR	29.4/42.5/51.2	88.2 /93.9/ 97.8	62.8/ 78.8/91.8

Table 1: Visual localization performance comparison using SIFT descriptors across varying dimensions on Aachen Day/Night and InLoc datasets. Results show localization success rates at 0.25m/2°, 0.5m/5°, and 5m/10° error thresholds (format: threshold1/threshold2/threshold3). GFR often matches or exceeds PCA, DL-US, and K-PCA across many dimensional reductions, with highlighted values indicating best performance in each category. Baseline SIFT-128 represents full-dimensional descriptor performance.

Aachen Day-Night Results. Table 1 shows that at 64 dimensions, GFR attains 88.2% day-time accuracy (0.25m/2°), comparable to or above PCA (86.2%), DL-US (86.7%), and K-PCA (87.0%). Night-time performance remains competitive at 62.8% while halving dimensionality. At 32 dimensions, GFR achieves 87.8%/93.5%/97.3% day accuracy across thresholds, often matching or exceeding competing approaches.

For HardNet (Table 2), GFR-64 achieves (88.3%, 94.9%, 98.2%) day-time accuracy, closely matching the HardNet-128 baseline (87.9%, 94.6%, 98.3%) at half the dimensionality.

InLoc Indoor Results. GFR performs competitively across dimensions. For SIFT at 64D, GFR reaches (29.4%, 42.5%, 51.2%). HardNet descriptors reach (34.3%, 52.9%, 61.6%). At moderate dimensions (32–64), GFR often matches or exceeds full-dimensional performance, consistent with a denoising effect. Figure 3 provides qualitative examples of successful localization using GFR-compressed descriptors.

Dim	Method	InLoc	Aachen Day	Aachen Night
128	HardNet	34.2/50.0/61.5	87.9/94.6/98.3	63.3/80.4/91.8
16	PCA	18.3/ 30.1/36.1	83.8/ 90.7 /95.2	26.9/34.2/45.6
	DL-US	16.3/26.3/32.1	83.3/90.2/94.1	29.5/33.7/44.6
	K-PCA	17.3/26.0/32.1	83.2/89.8/95.0	28.0/34.7/42.0
	GFR	18.7 /28.6/34.3	84.2 /90.4/ 95.8	30.1/35.7/47.1
24	PCA	25.4/ 39.0 /45.5	86.0/92.5/ 97.3	44.1/54.5/68.9
	DL-US	23.1/33.7/43.8	85.5/93.1/96.2	40.4/ 58.0 /68.9
	K-PCA	26.1 /36.7/42.6	86.5 /92.7/96.6	45.6/56.5/65.8
	GFR	23.4/36.7/ 45.6	86.4/ 93.2 /96.8	48.2/58.0/70.5
32	PCA	30.0/43.6/51.7	87.1/93.2/ 97.9	54.5/68.9 /81.9
	DL-US	27.3/42.6/52.4	87.0/93.5/97.7	53.4/64.7/80.4
	K-PCA	29.0/43.0/51.8	87.7 /93.9/97.3	53.4/67.4/78.8
	GFR	30.5/46.0/53.6	87.4/ 94.1/97.9	54.5 /68.4/ 84.0
64	PCA	32.7/50.1/60.2	87.8/94.6/ 98.3	60.1/77.7/ 92.3
	DL-US	36.1 /50.7/61.0	87.9/94.5/98.2	57.0/ 79.3 /91.2
	K-PCA	33.3/48.9/60.6	88.2/94.3/98.2	60.7/78.8/90.2
	GFR	34.3/ 52.9/61.6	88.3/94.9 /98.2	62.2 /78.3/91.2

Table 2: HardNet descriptor dimensionality reduction impact on visual localization across Aachen Day/Night and InLoc datasets. Localization success rates are reported using three error tolerance criteria: 0.25m/2°, 0.5m/5°, and 5m/10° (format: strict/moderate/relaxed thresholds). GFR maintains competitive performance with the original HardNet-128 baseline while often outperforming alternative compression methods (PCA, DL-US, K-PCA); across some thresholds, winners are mixed.

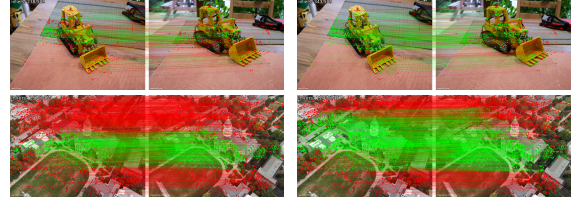


Figure 3: Visual localization results using SIFT-PCA-64 and SIFT-GFR-64 on our custom datasets (Kid Toy and WAMI Jesse Hall). For each image pair, the left image is the query, and the right is the retrieved database image with the most inlier matches, as computed by PnP+RANSAC. Red lines indicate incorrect feature matches, and red boxes denote failed localizations. Green lines and boxes represent correct matches and successful localizations. Best viewed in color and zoomed in. These internal datasets qualitatively illustrate matching/localization behavior; no quantitative claims are derived from them.

3.3 HPatches Evaluation: Matching, Retrieval, and Verification

We evaluate patch-level performance on 116 HPatches sequences across three tasks.

Image Matching. Figure 4 shows GFR achieves competitive mAP across dimensions, with strong performance at 64D. GFR maintains consistent results across Easy, Hard, and Tough scenarios, indicating

orthogonal decomposition preserves discriminative information for feature correspondence.

Patch Retrieval. Figure 5 demonstrates GFR’s competitive retrieval performance, often matching or exceeding SIFT-128 baseline. At 32-64 dimensions, GFR achieves good accuracy with substantial computational savings.

Patch Verification. Figure 6 shows GFR maintains competitive performance across dimensions. While verification shows balanced performance distribution among methods, GFR demonstrates consistent advantages at moderate dimensions (32-64).

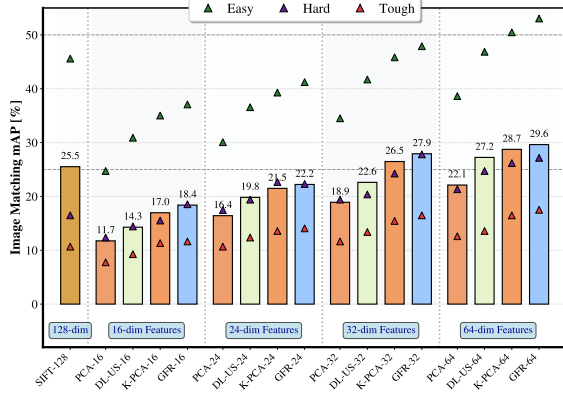


Figure 4: Image matching performance evaluation measuring mean Average Precision (mAP) across varying descriptor dimensions. GFR often matches or exceeds traditional dimensionality reduction methods (PCA variants, K-PCA) and deep learning approaches (DL-US), with strong results at 64 dimensions. Performance metrics are stratified by matching difficulty: Easy, Hard, and Tough scenarios (scatter points).

3.4 Theoretical Validation and Empirical Insights

Our results validate advantages of iterative orthogonalization over global variance maximization, revealing key insights about discriminative preservation.

Denosing Effect. GFR-reduced descriptors often outperform full-dimensional counterparts, supporting our hypothesis. SIFT-GFR-64 achieves 88.2% vs 87.2% for SIFT-128, while HardNet-GFR-64 matches HardNet-128 (88.3% vs 87.9%). This suggests iterative orthogonalization filters redundant information while preserving discriminative components.

Descriptor Paradigm Analysis. Hand-crafted descriptors (SIFT) show larger improvements (2-5%) than learned descriptors (HardNet, 1-3%), validating that learned descriptors contain less exploitable redundancy due to training optimization.

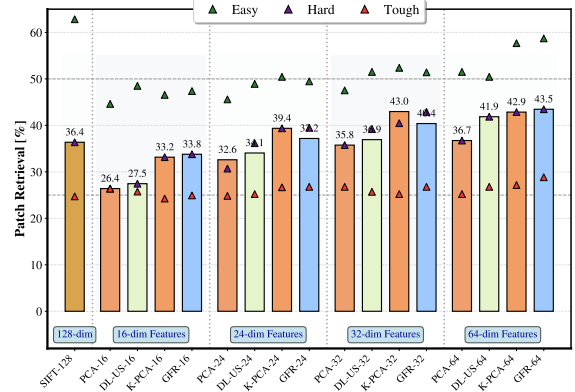


Figure 5: Patch retrieval accuracy comparison demonstrating GFR’s effectiveness in content-based image retrieval tasks. The proposed method maintains robust performance across all dimensional reductions (16, 32, 64), often exceeding or matching baseline SIFT-128 and competing approaches. Evaluation encompasses three complexity tiers (Easy, Hard, Tough) represented by individual data points.

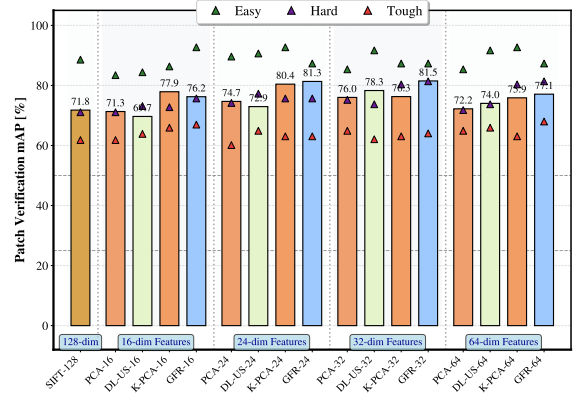


Figure 6: Patch verification results showing classification performance using mean Average Precision metric. While GFR demonstrates competitive performance against established methods across dimensional scales, the verification task reveals more balanced performance distribution among approaches. Results are segmented by verification complexity levels (Easy, Hard, Tough conditions) shown as scatter points.

Task-Dependent Validation. The strongest GFR improvements appear in visual localization and image matching, which require precise spatial relationship preservation. Patch verification shows more modest improvements due to reduced sensitivity to fine-grained discriminative preservation.

Computational Efficiency. Reducing dimensions from 128 to 32-64 decreases matching complexity from $O(n^2 \times 128)$ to $O(n^2 \times 32-64)$, providing 2-4x speedup and 50-75% storage reduction.

Optimal Operating Points. Results identify 32-64 dimensions as optimal trade-offs. GFR-64

matches/exceeds full-dimensional performance while halving computational requirements; GFR-32 maintains competitive performance with substantial efficiency gains.

4 ABLATION STUDY

To validate the design choices of our Gram-Schmidt feature reduction (GFR) method, we conducted an ablation study focusing on the key hyperparameter: the function family used for orthogonalization. The computational cost of GFR is primarily determined by the complexity of the function family, $f(z)$, employed in the Gram-Schmidt process. Following the methodology in (Yaghoobi et al., 2025), we evaluated three function families: linear functions ($f_1(z) = z_i$ for $i \in [d]$), quadratic polynomial functions ($f_2(z) = z_i z_j$ for $i, j \in [d]$), and cubic polynomial functions ($f_3(z) = z_i z_j z_k$ for $i, j, k \in [d]$), where d denotes the number of features in the feature map.

Our analysis revealed that the linear function family (f_1) produces results similar to PCA, which does not provide optimal performance for our application. Conversely, the cubic function family (f_3) imposes significant computational overhead in terms of both time and resource consumption. As summarized in Table 3, the quadratic function family (f_2) achieves the optimal balance between computational efficiency and attribution quality. Therefore, we adopt $f_2(z)$ as our orthogonalization function.

Table 3: Computational cost comparison (minutes) for different orthogonalization functions in GFR.

Function Family	$f_1(z) = z_i$	$f_2(z) = z_i z_j$	$f_3(z) = z_i z_j z_k$
Detector+GFR($f(z)$)	0.81	1.1	5.21

5 Conclusion

We have introduced Gram-Schmidt Feature Reduction (GFR), a sequential orthogonalization framework that addresses fundamental limitations of variance-maximization approaches to descriptor compression. By iteratively identifying principal directions in residual variance and enforcing orthogonality in the sample-score space, GFR preserves complementary discriminative components that global optimization criteria can suppress, yielding compact representations that maintain or enhance performance across diverse computer vision tasks.

Our comprehensive evaluation across visual localization benchmarks (Aachen Day-Night, InLoc)

and patch-level tasks (HPatches) demonstrates that GFR-compressed descriptors at 16-64 dimensions consistently match or exceed their 128-dimensional counterparts. Notably, SIFT-GFR-64 achieves 88.2% localization accuracy compared to 87.2% for full-dimensional SIFT-128, while HardNet-GFR-64 matches the 87.9% baseline performance at half the computational cost. These results validate our hypothesis that dimensionality reduction can function as an effective denoising mechanism, filtering redundant information while retaining essential discriminative structure.

The practical advantages of GFR are substantial: reducing descriptor dimensionality from 128 to 32-64 dimensions yields 2-4 $\times$ matching speedup, 50-75% storage reduction, and 2-8 $\times$ memory savings. These efficiency gains enable deployment in resource-constrained environments including mobile robotics, real-time visual localization systems, and large-scale image retrieval applications where computational and memory constraints are critical design considerations.

Our analysis reveals important insights into descriptor compression across paradigms. Hand-crafted descriptors (SIFT) exhibit larger improvements (2-5%) under GFR compression compared to learned descriptors (HardNet, 1-3%), suggesting that learned representations contain less exploitable redundancy due to end-to-end training optimization. Task-dependent analysis shows that localization and matching—which require precise preservation of spatial relationships—benefit most from GFR’s discriminative preservation, while verification tasks show more modest gains due to reduced sensitivity to fine-grained discriminative information.

Critically, GFR requires no descriptor-specific training and generalizes across both hand-crafted and learned descriptor paradigms. This training-free property contrasts sharply with recent deep learning approaches that require expensive retraining when deployment conditions change, making GFR particularly practical for operational computer vision systems. The method’s computational complexity during projection learning scales linearly with descriptor dimensionality, and once learned, the projection matrix enables efficient real-time compression.

Our work establishes several directions for future research. First, extending GFR to incorporate task-specific discriminative objectives could further enhance performance on specialized applications. Second, investigating adaptive dimensionality selection strategies that automatically determine optimal compression rates based on descriptor statistics and task requirements could improve deployment flexibility. Third, combining GFR with quantization techniques

could yield additional storage and computational benefits for large-scale retrieval systems. Finally, exploring GFR’s applicability to emerging learned descriptors beyond HardNet—including transformer-based and self-supervised representations—would validate its generalization to modern deep learning architectures.

In conclusion, Gram-Schmidt Feature Reduction provides a principled, efficient, and practical solution to descriptor compression that preserves discriminative information while enabling deployment in resource-constrained scenarios. By demonstrating that reduced-dimensional descriptors can match or exceed full-dimensional performance, our work challenges conventional assumptions about the relationship between descriptor dimensionality and task accuracy, opening new avenues for efficient computer vision system design.

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